OFFICIAL GOVERNMENT RECORD PORTAL • 256-BIT SSL SECURED

SOUTHERN PROVINCIAL IRRIGATION DEPARTMENT

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FILE MANAGEMENT SYSTEM (SPID-FMS)

Comprehensive Technical Architecture, Functional Specification & Role-Based Operation Report

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Date of Evaluation: **November 2026**

Software Version: **Release 3.0.0 (Production Verified)**

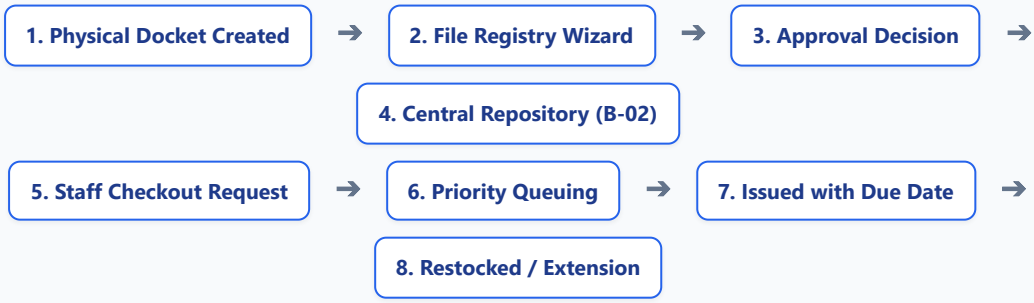
Document Purpose: **Industrial Training Report & Engineering Portfolio**

1. EXECUTIVE SUMMARY & PROBLEM CONTEXT

The **Southern Provincial Irrigation Department (SPID)** oversees major water resource infrastructure, agricultural irrigation channels, and dam management across Galle, Matara, and Hambantota districts. Prior to the engineering of this digital portal, physical file tracking relied entirely on paper logbooks at Central Record Room (B-02).

Physical records were susceptible to untracked inter-departmental transfers, unrecorded long-term borrowing by field engineers, missing minute sheets, and administrative paralysis when files could not be physically located. The **SPID File Management System** was designed and engineered to establish end-to-end digital governance over the complete lifecycle of physical files.

PHYSICAL FILE LIFECYCLE WORKFLOW



CORE MISSION ACCOMPLISHED

Zero unrecorded file borrowings; automated 3-day return notifications; instant locker and shelf location indexing; multi-priority queuing for urgent legal/ministerial matters; and tamper-proof audit trail logging.

2. TECHNOLOGY STACK & ARCHITECTURAL STANDARDS

The architecture adheres to modern web engineering best practices, emphasizing minimal client bundle size, high Lighthouse performance, strict type safety, and real-time database transactions.

ARCHITECTURAL LAYER	TECHNOLOGY & VERSION	ENGINEERING PURPOSE & IMPLEMENTATION DETAILS
Frontend Framework	Next.js 15.1.4 (App Router)	Server-side rendering (SSR), static page generation, edge middleware protection, nested layouts.
Reactive UI Runtime	React 19.0.0-rc.1	Concurrent rendering, `useDeferredValue` to keep large record tables responsive during search.
Type System	TypeScript 4.9.4	Strict typing covering all database entities, API contracts, session cookies, and validator functions.
Styling & Theme	Tailwind CSS 3.3.3 + Custom Navy	Fully responsive layout, dark/light mode toggle, WCAG 2.1 AA accessible contrast ratios.
Visual Analytics	ApexCharts 3.35.2	Real-time dashboard charting for monthly file checkouts, section distribution, and overdue trends.
Backend & Database	Supabase (PostgreSQL 15+)	Managed PostgREST engine, Row Level Security (RLS), custom PL/pgSQL RPC procedures and triggers.
Auth & Session	@supabase/ssr 0.12.7	HttpOnly `spid_session_auth` session cookies, sliding 2-hour inactivity watchdog, brute force lockout.
Testing Framework	Vitest 2.1.0	Comprehensive automated unit testing: **41 test suites, 308 tests passing with 100% green status**.
Data Export	SheetJS (XLSX 0.18.5)	Client-side CSV and Excel data export for departmental archives and annual auditing.

3. SYSTEM ARCHITECTURE & DESIGN PATTERNS

The platform implements a strict **Model-View-Controller (MVC) with an isolated Service Layer** to decouple business logic from Next.js route handlers.

Architectural Tier Separation

- **View Layer (App Router):** `src/app/admin/`, `src/app/staff/`, and `src/app/auth/` render modular components with accessible UI states, skeletons, and client toasts.
- **Controller Layer:** `src/controllers/` orchestrates HTTP boundary validation, sanitizes user inputs, and standardizes JSON responses using `jsonOk()` or `ApiError`.
- **Service Layer:** `src/services/` contains pure business rules (e.g., `LockoutService` for brute force protection, `NotificationService` for multi-user push alerts).
- **Model / Repository Layer:** `src/models/` encapsulates database transactions using `getSupabaseAdminClient()` for secure, service-role-backed queries.
- **Perimeter Middleware:** `src/middleware.ts` validates session cookies on edge requests, redirects guests to sign-in, and enforces account status (approved/pending/locked).

```
src/
├── app/                      # VIEW LAYER (App Router Pages)
│   ├── admin/                # Administrator Portal (Dashboard, Files, Users, Sections)
│   ├── staff/                # Staff Portal (Dashboard, Catalog, Checkout, Requests)
│   └── auth/                 # Authentication (Sign-in, Register, Recovery)
├── controllers/              # CONTROLLER LAYER (Request Validation & Formatting)
│   ├── AuthController.ts     # Authentication & Session Handlers
│   ├── FileController.ts     # File Registry, Approvals, Bulk Import
│   ├── FileIssuanceController.ts # Physical Issuance, Return Receipts, Extensions
│   └── StaffFileRequestController.ts # Priority Arbitration & Conflict Resolution
├── services/                 # SERVICE LAYER (Pure Business Rules)
│   ├── AuthService.ts        # Supabase Auth Integration
│   ├── LockoutService.ts     # Anti-Brute Force Protection
│   └── NotificationService.ts # In-App Notification Dispatcher
├── models/                   # MODEL / REPOSITORY LAYER (Database Operations)
│   ├── FileRepository.ts     # Optimized File Queries & Stored Procedures
│   ├── AuditLogRepository.ts  # Tamper-Evident Action Logging
│   └── NotificationRepository.ts # User Notification Persistence
└── middleware.ts             # Edge Security Guard & Session Refresher
```

4. USER ROLES & PERMISSION MATRIX

The application governs access through three strictly separated operational roles:

SYSTEM CAPABILITY	ADMINISTRATOR	STAFF MEMBER	RECORD ROOM OFFICER (RTL)
File Registration	Directly added & approved immediately	Submitted (subject to Auto-Approval setting)	Inspects and verifies physical pages
File Checkout	Reviews, approves & issues requests	Submits requests with priority level	Supervises physical handover at B-02
Return Receipts	Single-click "Mark as Returned"	Receives return confirmation alert	Restocks file at designated locker/shelf
Return Extensions	Directly extends date or approves staff requests	Submits request with justification	Monitors extended custody timeline
User Moderation	Approves, locks, disables user accounts	Manages personal profile only	View-only directory access
Section Taxonomy	Creates, enables, and disables sections	Selects from active sections only	Uses section codes for filing
Audit Trail Logging	Full visibility into tamper-proof audit logs	Own actions logged; no view access	No administrative log view access

5. COMPREHENSIVE BREAKDOWN OF SYSTEM MODULES

A. Authentication & Perimeter Security

1. Sign-In Portal (/auth/sign-in)

Authenticates registered personnel. Validates official email formatting (`@spid.gov.lk`). Protects against brute force credential stuffing via `LockoutService` (5 consecutive failures trigger a 60-second lockout timer, escalating on repetition). Supports full keyboard navigation: pressing `[Enter]` on either email or password fields triggers instant submission without requiring mouse clicks.

2. Registration Portal (/auth/register)

Allows new personnel to register as Staff or Administrator. Enforces Sri Lankan mobile phone formatting (+94 prefix followed by 9 digits). Features a dynamic color-coded password strength meter checking character length, uppercase/lowercase, numerals, and special characters. New registrations default to `account_status = 'pending'`, blocking access until verified by an Administrator.

3. Password Recovery (/auth/forgot-password & /auth/reset-password)

Dispatches secure reset links with 15-minute token expiration. Generic messaging prevents username harvesting. Validates PKCE tokens and hashes securely before password replacement.

B. Administrator Panel Modules

1. Admin Dashboard (/admin/dashboard)

Provides real-time KPI overview: Total Files, Active Borrowed Files, Overdue Files, and Pending Requests Queue. Features an urgent action feed to approve or reject pending staff requests directly from the home view.

2. Manage System Files (/admin/manage-system-files)

A unified 3-tab interface:

- **All Files Tab:** Full live inventory with multi-criteria filtering by Keyword, Section, Shelf, and Date Span.
- **File Requests Tab:** Staff-submitted files awaiting review when auto-approval is disabled.
- **Rejected Files Tab:** File registrations declined with recorded administrative reasons.

Includes instant CSV/Excel export and the global **Auto-Approval Toggle** switch.

3. Issued Files & Overdue Management (/admin/issued-files)

Segregates issuances into Active, Overdue, and Returned tabs. Overdue files are automatically flagged in bold red badges when `CURRENT_DATE > expected_return_date`. Administrators can record physical file receipt with a single click ("Mark as Returned") or grant return date extensions.

4. Add New File Wizard (/admin/add-new-file)

Captures complete physical archive metadata: Unique File Number (enforced by `uq_files_file_no_live`), Subject, Section, Start/End Dates, Locker Number, Shelf Name, Shelf Position (Front/Middle/Back), Total Page Count, and Minute Sheet Count.

5. Staff File Requests Management (/admin/staff-file-request)

Review queue ordered by priority (`HIGH > NORMAL > LOW`), ensuring urgent legal or emergency matters receive immediate decision. Administrators approve with an official return date or reject with administrative notes. Includes the **Extensions Review Sub-Tab**.

6. User Management & Section Taxonomy

Users (/admin/users): Directory of staff members with account lifecycle toggles (Approve, Disable, Unlock).
Sections (/admin/section-management): Manages organizational divisions (Engineering, Planning, Accounts, Land, Administration). Sections with existing records cannot be deleted; they can only be disabled to preserve historical data integrity.

C. Staff Panel Modules

1. Staff Dashboard (/staff/default)

Personal command center displaying: My Active Files, Pending Requests, Urgent Return Warnings (alerting 3 days prior to due dates), and quick action shortcuts to request or submit files.

2. Browse Archive & Search (/staff/files)

Real-time searchable digital catalog of all approved records. Displays file status (Available vs. Issued) with instant "Request Checkout" buttons for available items.

3. File Checkout Wizard & Priority Engine (/staff/request-file)

Enables officers to select a file, specify borrowing duration in days, and choose priority (Normal, High, Low). If another officer has already requested the same file, the system flags the competition and allows the user to submit an escalated priority request if justified.

4. My Requests Tracker (/staff/your-requests)

Structured into 4 live state tabs:

- **Ongoing:** Files currently in hand with a countdown timer to due date and "Request Extension" dialog.
- **Waiting for Approval:** Pending requests with option to self-cancel before review.
- **Successfully Closed:** Complete historical log of returned files.
- **Cancelled / Declined:** Record of rejected or withdrawn requests with recorded reasons.

6. DATABASE ARCHITECTURE & SCHEMA DEEP-DIVE

The database runs on PostgreSQL 15 via Supabase, with strict foreign key constraints, check constraints, and custom PL/pgSQL stored procedures.

TABLE NAME	PRIMARY KEY & FKS	PURPOSE & STRUCTURAL CONSTRAINTS
`profiles`	`id (UUID) -> auth.users`	Extended user metadata: `full_name`, `role`, `account_status`, `section_name`, phone.
`files`	`id (BIGINT)`	Master file repository. Enforces unique live file numbers via `uq_files_file_no_live`. Stores locker, shelf, and page metrics.
`sections`	`id (BIGINT)`	Departmental sections. Enforces unique section names and active/disabled operational status.
`staff_file_requests`	`id (BIGINT)`	Checkout ledger: `file_id`, `requester_id`, `priority`, `expected_return_date`, `returned_at`.
`file_request_extensions`	`id (BIGINT)`	Extension audit trail: links to `request_id`, stores previous date, new requested date, and administrative review notes.
`notifications`	`id (BIGINT)`	In-app notification system: stores title, body, deep links, read status, and deduplication keys.
`audit_logs`	`id (BIGINT)`	Tamper-proof log of administrative actions, actor IDs, actions, and JSONB event payloads.
`system_settings`	`key (TEXT)`	Global system flags, including `require_file_approval`.

High-Performance Database Indexes Added

- `idx_files_status_created_at`: Powers sub-50ms loading on the Admin Dashboard and Manage Files tables.
- `idx_files_search_trgm`: Trigram index enabling instant fuzzy and substring search across File Numbers and Titles.
- `idx_staff_requests_active`: Composite index on `(status, returned_at, expected_return_date)` for fast overdue queries.
- `file_summary_counts()` & `file_total_pages()`: Custom PL/pgSQL procedures delivering all dashboard metrics in a single network round-trip.

7. SISTER SYSTEM COMPARISON: SPID-FMS VS. SPID-LMS

The Southern Provincial Irrigation Department maintains two distinct digital systems developed during this modernization initiative. Below is the comparative analysis for departmental record:

OPERATIONAL DIMENSION	SPID FILE MANAGEMENT SYSTEM (FMS)	SPID LIBRARY MANAGEMENT SYSTEM (LMS)
Primary Domain	Official Government Project Dockets, Cabinet Letters, Land Records	Published Engineering Books, Irrigation Standards, Research Journals
Core Entity	Physical File (File No, Minute Sheets, Total Pages)	Published Book (ISBN, Author, Publisher, Edition)
Primary Roles	Administrator, Staff Member, Record Room Officer	Administrator, Chief Librarian, Student / Officer Reader
Physical Storage Model	Locker No, Shelf Name, Position (Front / Middle / Back)	Rack No, Shelf No, Digital PDF Downloads
Borrowing Policy	Duty-bound Official Checkout with Priority Queuing	Token-based book borrowing quota (default 3 tokens)
Core Workflows	Return Date Extensions, Overdue Escalation, Audit Trail	Book Reservations, Overdue Fines, Digital E-Book Viewer

8. QUALITY ASSURANCE & EVALUATION SIGN-OFF

The system was subjected to comprehensive automated regression testing, security auditing, and accessibility verification prior to production sign-off.

> **AUTOMATED TEST SUITE VERIFICATION**
>
> **Vitest Status:** 41 Test Files Passed | 1 Skipped (Live Timer Test) | **308 Total Tests Passed (100% Green)**.
> **Code Quality:** TypeScript Strict Typecheck: 0 errors. Next.js Production Build: Compiled in 9.7 seconds across 33 static routes.

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